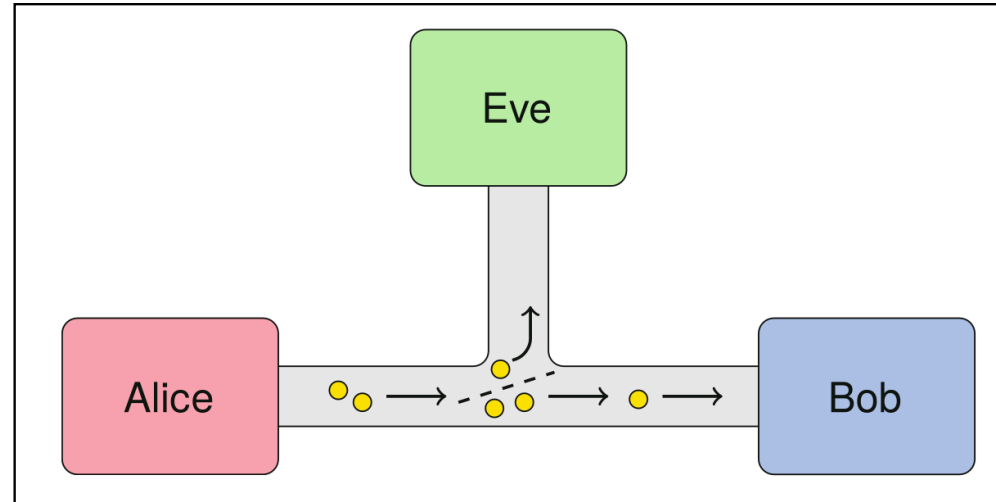


Photon-Number-Splitting (PNS) Attack



1. Ideal QKD uses single photons for qubits (e.g., polarization states).
2. In practice, weak laser pulses often emit multiple photons.
3. **PNS Attack:** Eve intercepts and keeps one photon from multi-photon pulses, waits for Alice's basis announcement, then measures stored photons in correct basis.
4. This lets Eve gain perfect knowledge of the key without detection.
5. Advanced BB84 QKD to overcome PNS Attack: **Decoy State QKD.**





Decoy State QKD

1. **WCP Photon Distribution:** $P(n) = \frac{\mu^n e^{-\mu}}{n!}$
2. Eve can perform **Quantum Non Demolition Measurement** to learn n without disturbing encoded qubit.
 1. If $n = 1$: Then Eve blocks or forwards depending on attack strategy
 2. If $n \geq 2$: Eve removes one photon and forwards the remaining photons to Bob
 3. When Alice and Bob publicly announce their measurement bases, Eve measures the stored photon in correct basis and obtained the transmitted bits without introducing any detectable errors
3. **Decoy State Method:** Alice varies the intensity of each transmitted pulse:
 1. **Signal state:** intensity μ (used to generate the actual key)
 2. **Decoy state(s):** intensity ν (used only for parameter estimation)
 3. **Vacuum state** (optional but standard): intensity 0
4. **Crucial Observation:** A single-photon pulse is physically identical regardless of whether it originates from a signal state or decoy state.
5. Therefore, Eve cannot distinguish whether a detected single photon belongs to a signal pulse or a decoy pulse.



Decoy-State BB84 Protocol

1. **Step 1: State Preparation:** For every transmitted pulse, Alice randomly selects:
 1. A pulse intensity (signal, weak decoy, or vacuum),
 2. A basis (Z or X), and a bit value (0 or 1).
 3. The bit is encoded into one of the BB84 polarization states.
2. **Step 2: Quantum Transmission & Measurement:** Alice transmits the encoded WCP through the quantum channel. Bob independently chooses a measurement basis and records whether a detection event occurs.
3. **Step 3: Basis Reconciliation:** Using authenticated channel, Alice & Bob reveal only their basis choices. Bits corresponding to different bases are discarded, producing the sifted key.
4. **Step 4: Intensity Announcement:** After basis reconciliation, Alice announces which pulses were signal states and which were decoy states. Bob classifies his detection events according to the transmitted intensity.
5. **Step 5: Parameter Estimation:** For each intensity level, Alice and Bob estimate the
Gain: $Q_\mu = \frac{\text{Detected pulses}}{\text{Transmitted pulses}}$ & **QBER:** $E_\mu = \frac{\text{Number of erroneous bits}}{\text{Detected bits}}$



Decoy-State BB84 SKR Analysis

1. **Gain:** $Q_\mu = \sum_{n=0}^{\infty} P_\mu(n) Y_n$, where Y_n is the **yield**, i.e., the probability that Bob detects an n -photon pulse.
2. Similarly, $E_\mu Q_\mu = \sum_{n=0}^{\infty} P_\mu(n) Y_n e_n$, where e_n is the error probability associated with an n -photon pulse.
3. **Security Analysis:** The security of decoy-state QKD relies on the fact that Eve cannot distinguish whether a given n -photon pulse originates from a signal state or a decoy state.
4. Consequently, the yield and error rate depend only on the photon number, $Y_n^{(\text{signal})} = Y_n^{(\text{decoy})}$ and $e_n^{(\text{signal})} = e_n^{(\text{decoy})}$
5. **SKR:** $R = q \left[Q_1 (1 - H_2(e_1)) - Q_\mu f(E_\mu) H_2(E_\mu) \right]$,
where , q is the sifting factor, Q_μ is the signal-state gain, $Q_1 = P_\mu(1)Y_1$ is the single-photon gain,
- E_μ is the observed QBER, , e_1 is the single-photon error rate,
- $f(E_\mu)$ is the error-correction efficiency, $H_2(x)$ is the binary entropy function.
6. **Step 6:** Error Correction and Privacy Amplification

Decoy-State BB84 Protocol Security



1. Suppose Eve attempts a PNS attack by blocking all single-photon pulses and forwarding only multi-photon pulses.
2. Decoy and signal states have different photon-number distributions due to their different mean photon numbers.
3. A PNS attack **alters the expected detection statistics** (gain and QBER) of signal and decoy states.
4. During **parameter estimation**, Alice and Bob compare the observed statistics with the theoretical Poisson predictions.
5. Any inconsistency reveals Eve's presence, making the PNS attack detectable.
6. **Result:** Decoy-state QKD effectively defeats PNS attacks while using practical weak coherent pulse sources.

Advantages of Decoy State QKD

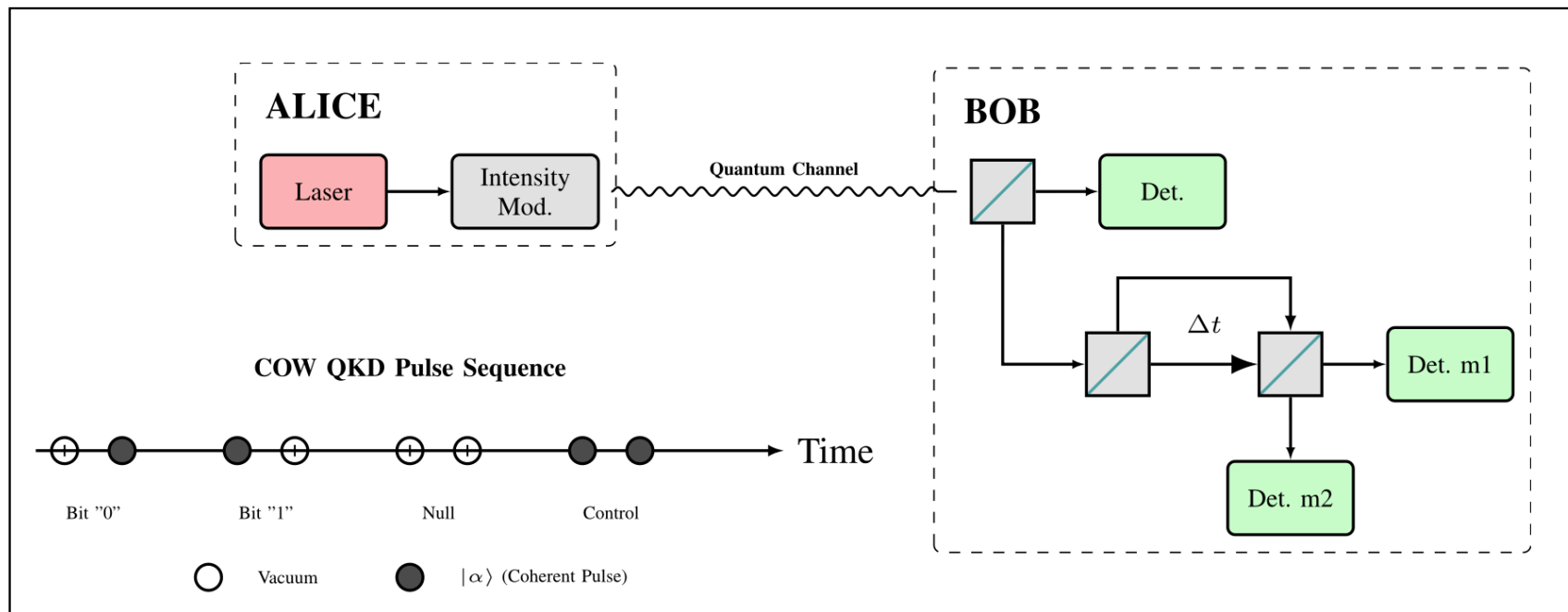


1. Provides unconditional security using inexpensive weak coherent laser sources.
2. Effectively defeats PNS attacks.
3. Significantly increases the achievable transmission distance.
4. Produces higher secret key rates than standard weak-coherent BB84.
5. Requires only an additional **intensity modulator**, making it easy to integrate into existing BB84 systems.
6. Forms the basis of nearly all modern commercial discrete-variable QKD systems.

Coherent One-Way (COW) QKD Protocol



1. The protocol encodes information using pairs of time slots with either vacuum states or coherent pulses ($|\alpha\rangle$)
2. The bit value '0' is encoded as ($|0\rangle, |\alpha\rangle$) and bit '1' as ($|\alpha\rangle, |0\rangle$).
3. Furthermore, ($|0\rangle, |0\rangle$) represents a null sequence, and ($|\alpha\rangle, |\alpha\rangle$) is used as a control sequence to check the interferometric phase reference.



COW QKD Protocol



1. Advantages:

1. Very simple transmitter (only intensity modulation).
2. No random basis switching.
3. High secret-key generation rates.
4. Excellent for long-distance fiber communication.
5. Robust against polarization fluctuations.
6. Compatible with standard telecom components.

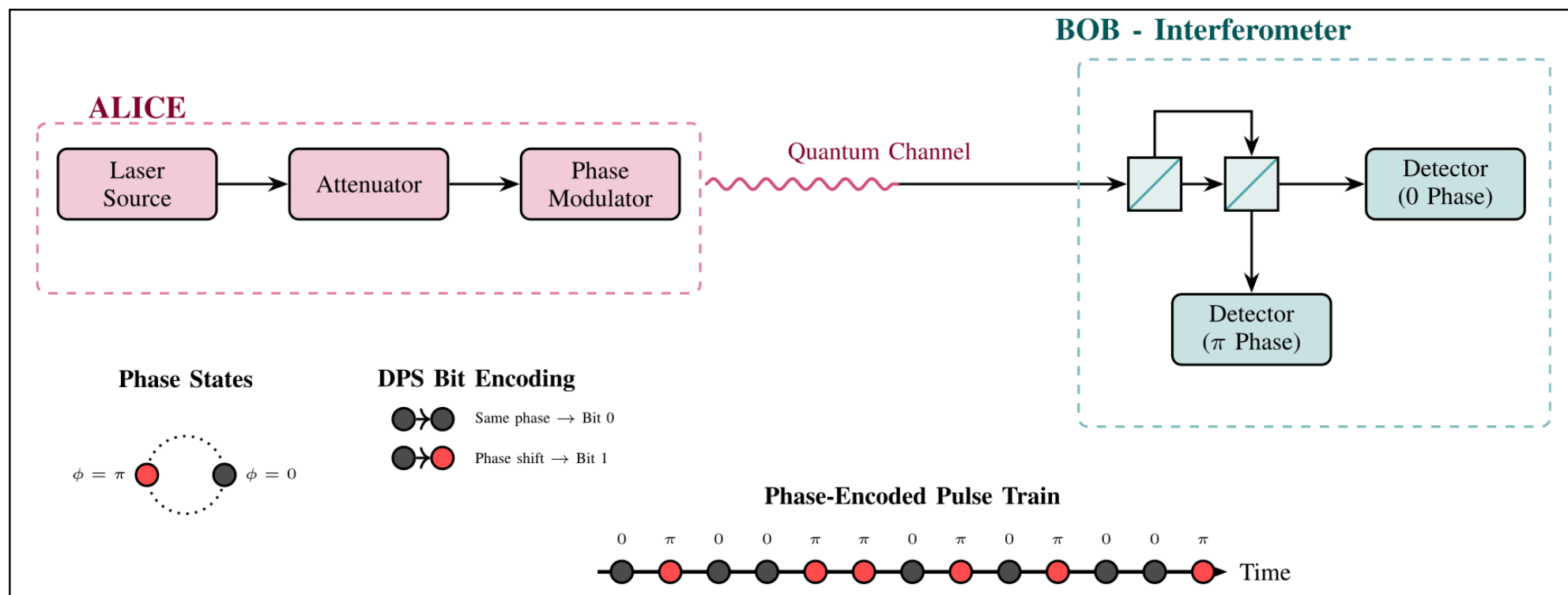
2. Limitations:

1. Security proofs are more complex than those for BB84 and often rely on specific attack models, although significant progress has been made toward general security analyses.
2. Requires stable interferometer operation at the receiver.
3. Sensitive to coherence loss caused by laser phase noise or environmental fluctuations.
4. Interference visibility must remain high for reliable eavesdropping detection.

Differential Phase Shift (DPS) QKD Protocol



1. In DPS QKD, Alice sends a sequence of coherent light pulses with random phases (0 or π).
2. The key is encoded in the phase difference between consecutive pulses: same phase = 0 , different phase = 1 .
3. Bob measures these differences using a delay interferometer, where pulses interfere with the next one.
4. Security comes from the fact that individual photons are hard to measure without disturbing the sequence.



DPS QKD



1. Advantages:

1. A Simple transmitter and receiver architecture.
2. No active basis switching.
3. No basis reconciliation (higher sifting efficiency than BB84).
4. Well suited for high-speed fiber-optic communication.
5. Robust against polarization drift because information is encoded in phase.

2. Limitations:

1. Security proofs are more involved than those of BB84, and unconditional security for practical coherent-state implementations required significant later work.
2. Requires a highly stable interferometer.
3. Sensitive to phase noise and timing synchronization errors.
4. Performance degrades if the interference visibility is low.

Comparison of PM DV-QKD Protocols



Feature	BB84	DPS-QKD	COW-QKD
Information encoding	Polarization, phase, or time-bin states	Relative phase between adjacent pulses	Position of a pulse within a pair of time bins
Security monitoring	QBER	Interference of adjacent pulses	Interference visibility of adjacent pulses
Basis switching	Required	Not required	Not required
Receiver	Random basis measurement	One-bit-delay interferometer	Data detector + monitoring interferometer
Suitable medium	Fiber and free space	Fiber	Fiber
Transmitter complexity	Moderate	Simple	Very simple

Overview



Entanglement Based QKD Protocols



Why Use Entanglement in QKD?

1. Entangled photons can be used by Alice and Bob to establish a **secret key**.
2. Helps defend against the **Photon Number Splitting (PNS)** attack.
 1. PNS attack is less effective because producing two entangled photon pairs simultaneously is very unlikely.
3. Entangled states can be distributed by:
 1. Alice herself
 2. A third party (commonly called **Charlie**)
 3. Even **Eve**, the eavesdropper
4. The source is considered **untrusted**; we assume **Eve controls it** (worst-case scenario).

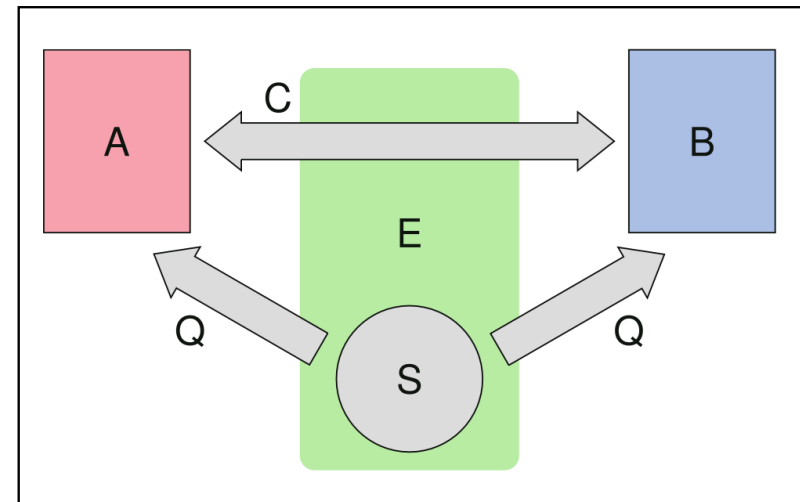
Protocol Setup and Security Features



1. Alice and Bob use:
 1. A **quantum channel** from the source to receive entangled states.
 2. An **authenticated classical channel** to communicate (Eve can listen, but not alter messages).
2. Security Analysis:
 1. Easier compared to prepare-and-send protocols.
 2. No direct quantum channel needed between Alice and Bob.

The setting of an entanglement-based protocol:

1. Alice (A) and Bob (B) receive entangled states via quantum channels (Q).
2. Eve (E) can listen to the classical channel (C) but cannot interfere.





The Ekert91 Protocol

1. Developed by Arthur Ekert in 1991 using **maximally entangled qubit pairs**.
2. The source distributes **maximally entangled states** to Alice and Bob.
3. Example state used in Ekert91:

$$|\Psi^-\rangle_{AB} = \frac{1}{\sqrt{2}}(|01\rangle_{AB} - |10\rangle_{AB})$$

4. Alice and Bob perform measurements on their qubits:
 1. Randomly choose observables A_i and B_i from specific sets.
 2. Observables lie in the x-z plane of the Bloch sphere.

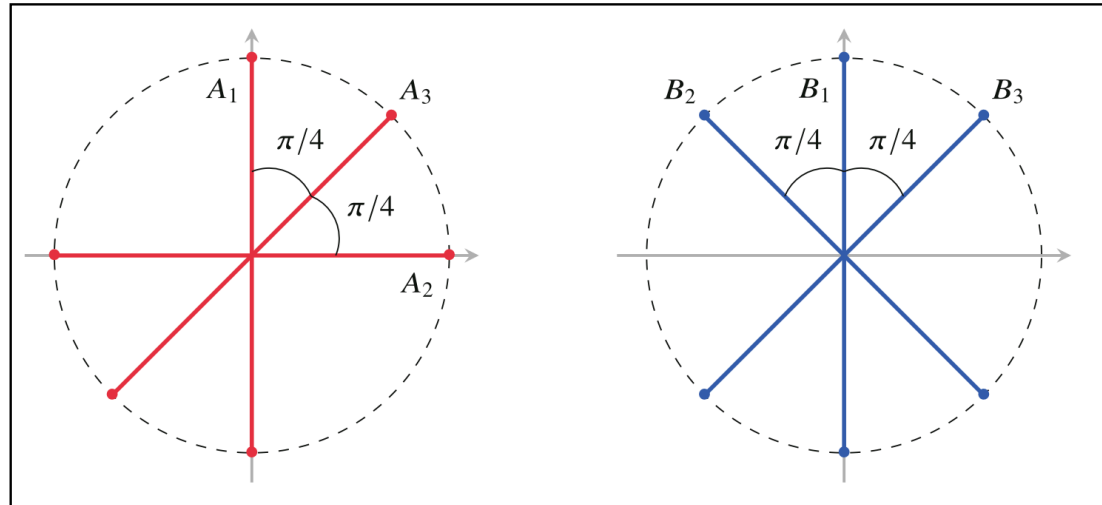
Measurement Settings in Ekert91

1. Alice and Bob randomly choose measurements from:

1. **Alice:** $A_1 = Z$, $A_2 = X$, $A_3 = \frac{1}{\sqrt{2}}(Z + X)$

2. **Bob:** $B_1 = Z$, $B_2 = \frac{1}{\sqrt{2}}(Z - X)$, $B_3 = \frac{1}{\sqrt{2}}(Z + X)$

2. These observables correspond to spin components in the $x - z$ plane.



Measurement directions for the Ekert protocol.

For Alice - left. For Bob - right.

Sifting and Security Steps



1. Alice and Bob share many entangled pairs.
2. Each randomly chooses one measurement from their respective sets.
3. They publicly announce the measurement bases (not results).
4. When bases match: (A_1, B_1) and (A_3, B_3) produce anti-correlated outcomes → **sifted key**.
5. Other basis combinations $(A_1B_3, A_1B_2, A_2B_3, A_2B_2)$ → used for **security check**.